Full Stack Assignment – 41

Assignment ID: FSD-41
Title: Multi-Level Review & Sign-Off System
Duration: 3 Hours

1️⃣ Objective

Design and develop a **Multi-Level Review & Sign-Off System** that allows submissions to go through multiple sequential review stages before final approval.
The system must strictly enforce review order and ensure **data persistence using a database**.

2️⃣ User Roles

SUBMITTER

- Register and log in to the system
- Create and submit items for review
- View submission status

REVIEWER_LEVEL_1

- Review submissions assigned to Level 1
- Approve or reject submissions

REVIEWER_LEVEL_2

- Review submissions approved by Level 1
- Approve or reject submissions

3️⃣ Functional Requirements

Authentication & Authorization

- User registration and login
- **JWT-based authentication**
- Protected backend APIs
- Strict **role-based access control**

Submission Management

- SUBMITTER can create submissions with required details
- Submissions must be stored in the database
- Review actions must update submission state persistently

🔄 Review Workflow

- REVIEWER_LEVEL_1 reviews first

- REVIEWER_LEVEL_2 reviews only after Level 1 approval
- Review history must be stored and retrievable

4️⃣ Business Rule (Critical)

⚠️ **Level 2 review must not occur unless Level 1 approval is completed.**

Any attempt to bypass review order must be rejected.

5️⃣ Technical Constraints (Mandatory)

- All submission and review data **must persist in a database**
- No hardcoded workflow logic
- JWT authentication and role validation mandatory
- Environment variables required

6️⃣ Frontend Requirements

- Role-based dashboards for SUBMITTER, REVIEWER_LEVEL_1, REVIEWER_LEVEL_2
- Submission creation view for SUBMITTER
- Review and decision view for reviewers
- Clear status indicators per review stage

UI polish not required; workflow correctness is key.

7️⃣ Submission Requirements

GitHub repository (frontend and backend)

Live frontend deployment link

Live backend deployment link

README.md must include:

- Project overview
- Tech stack
- User roles and permissions
- API endpoints
- Database schema
- Live deployment links

8️⃣ Evaluation Note

Evaluation focuses on **workflow sequencing, role enforcement, and data consistency**.